

Day 06 — IPV4

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THEORY

1. What is IPv4?

Internet protocol version 4 is a logical address used in layer 3 for logical addressing and routing for ip networks.

2. How many bits are in an IPv4 address?

32

3. What is an octet?

An ipv4 has 4 octets. An octet is an 8 bits group in binary.

4. How many octets are in an IPv4 address?

4

5. What is a subnet mask?

It tells devices which part of the ip address is for the network.

6. Difference between network portion and host portion?

Network portion is the same on the LAN, while the host portion is unique in each LAN

7. What is a private IPv4 address?

An ip address used inside private networks like home or office networks, not directly routed to the internet.

8. Write all three private IPv4 ranges.

10.0.0.0/8 → 10.0.0.0 – 10.255.255.255

172.16.0.0/12 → 172.16.0.0 – 172.31.255.255

192.168.0.0/16 → 192.168.0.0 – 192.168.255.255

9. What is a public IPv4 address?

It is the ip address used to communicate on the internet (public networks)

10. What is the IPv4 loopback address?

127.0.0.0/8

11. What is APIPA?

169.254.0.0/16, it may appear when an error occurs in network configuration or DHCP.

12. What does 169.254.x.x commonly indicate?

It is the APIPA, it indicates an error in dhcp/network configuration and is self-assigned by windows.

13. What is a default gateway?

It is the device that sends traffic to outside of a local network, most of the time it's the router.

14. Static vs dynamic IP addressing?

Static: manually assigned and kept consistent

Dynamic: assigned by DHCP.

15. What is DHCP?

Dynamic host configuration protocol; dynamically assigns ip address and other network configurations to devices as they connect to a network.

16. What is a network address?

When the host portion is zeros that represent the network address, they cannot be assigned to a host.

17. What is a broadcast address?

It is an address where the host portion of the address is all 1's, it cannot be assigned to a host.

18. Why is IPv4 knowledge important in cybersecurity?

Analysts work with source/destination IPs, firewall logs, server logs, packet captures, IDS/IPS alerts, SIEM events, Nmap results and Wireshark traffic.

PRACTICAL IDENTIFICATION

Your own: IPv4 address, Subnet mask, Default gateway, DNS server, DHCP status.

IPv4 address: 192.168.100.209

Subnet mask: 255.255.255.0

Default gateway: 192.168.100.1

DNS server: 8.8.8.8

DHCP status: enabled

SCREENSHOTS

1.ipconfig

```
Windows PowerShell
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PS C:\Windows\System32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::dd09:f6fe:ee7f:b4ac%22
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Unknown adapter McAfee VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 5:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::e3bb:c8ce:a438:c2e7%6
    IPv4 Address. . . . . : 192.168.100.209
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.100.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
PS C:\Windows\System32> |
```

2.ipconfig/all

```
PS C:\Windows\System32> ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : LAPTOP-AM22QKJ6
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-16
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::dd09:f6fe:ee7f:b4ac%22(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 789184551
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
NetBIOS over Tcpip. . . . . : Enabled
```

Unknown adapter McAfee VPN:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-74-4B-36-3C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 4:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #4
Physical Address. . . . . : C6-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 5:

```
Media State . . . . . : Media disconnected
```

Wireless LAN adapter Wi-Fi 5:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #5
Physical Address. : CA-13-75-3C-F1-01
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. : C4-13-75-3C-F1-01
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::e3bb:c8ce:a438:c2e7%5(Preferred)
IPv4 Address. : 192.168.100.209(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : Thursday, September 10, 2026 6:48:38 PM
Lease Expires : Saturday, September 12, 2026 3:20:24 PM
Default Gateway : 192.168.100.1
DHCP Server : 192.168.100.1
DHCPv6 IAID : 113513333
DHCPv6 Client DUID. : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
DNS Servers : fe80::1%5
8.8.8.8
8.8.4.4
fe80::f912:e97b:33cc:e623%5
fe80::1%5
NetBIOS over Tcpip. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : C4-13-75-3C-F1-02
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

3.Ping 127.0.0.1

```
PS C:\Windows\System32> ping 127.0.0.1
```

Pinging 127.0.0.1 with 32 bytes of data:

Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms